GE VERNOVA

ADMS

16.25.1

Annotation Importer/Exporter User Guide

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Change History

Date	Change Description
Jan 26, 2023	ADMS 3.16: • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
Apr 1, 2024	ADMS 16.7.0: • Updated document format and styling.

Contents

1. Introduction

Utility companies generally have Geographic Information System (GIS) asset management systems. These companies may want to share some annotations from these external systems and export them to ADMS.

The Annotation Importer/Exporter application is a tool that can be used to import and export annotations between binary SAV files and XML files, and to view and edit annotation binary files and XML files.

The Annotation Importer/Exporter operates between "annotation files" (binary dynamics files) and XML.

This application has the following capabilities:

- Saves, opens, and creates new annotation files
- Imports and exports XML to and from an annotation file

  If an XML file is imported, the contents are added to the current annotation file, in order to support multiple file input.

- Has a command-line interface, to allow for importing all XML files in a directory to a single annotation file

2. Using the Annotation Importer/Exporter Application

2.1. Starting the Annotation Importer/Exporter

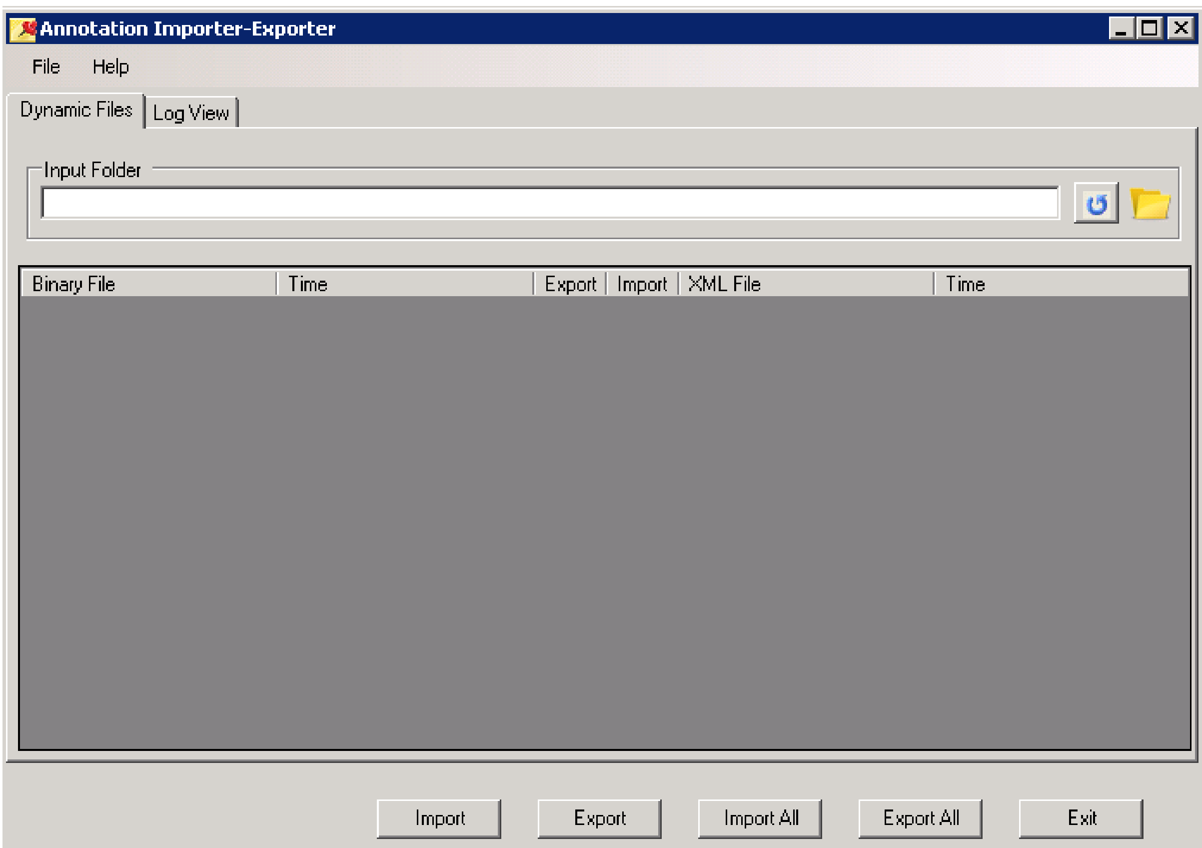
To start the Annotation Importer/Exporter application, perform the following steps:

1. Using Microsoft Windows Explorer, in the bin directory of the local installation, locate the `AnnotationImporter.exe` file.

   The default installation path is listed below:
   `D:\Eterra\Distribution\Tools\Modeling Tools\AnnotationImporter.exe`

2. To start the application, double-click the `AnnotationImporter.exe` file.

   The Annotation Importer/Exporter application starts, as shown below:



2.2. Importing Annotations from GIS Data

Annotations can be imported as XML-formatted data from an external source. These annotations

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are not considered to be part of the model, but they can be supplied by the GIS or any other external source.

To import annotations, perform the following steps:

1. On the Dynamic Files tab, click the Folder icon to browse to the location of the GIS data files, and then click OK.
2. Click the Refresh button .

   A list of all the XML files in the selected folder is displayed. Each listed XML file has a left arrow icon in the Import column.

3. To import a single file, select a file to import and click the Import button, or click the left arrow in the Import column.
4. To import all files, click the Import All button.

   The Log View tab contains messages relating to the import status of the file(s).

2.3. Exporting Annotations from Dynamic SAV Files

To export annotations, perform the following steps:

1. On the Dynamic Files tab, click the Folder icon to browse to the location of the dynamic SAV files.

   The files are located in the Dynamics folder where the configuration/model files are deployed. The default folder is ...\Distribution\Server\Dynamics.

2. Click the Refresh button.

   A list of all the SAV files in the selected folder is displayed. Each listed SAV file has a right arrow icon in the Export column.

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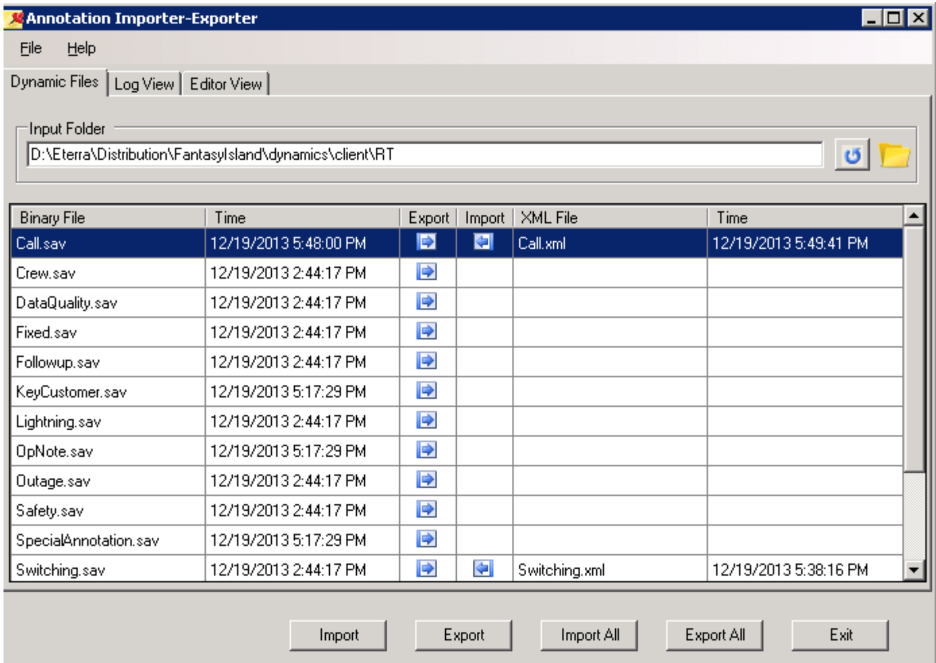

3. To export a single file, select a file to export and click the Export button, or click the right arrow  in the Export column.
4. To export all files, click the Export All button.

   The Log View tab contains messages relating to the export status of the file(s).

2.4. Opening, Viewing, and Editing an Annotation

To open and edit an annotation, perform the following steps:

1. Navigate to the Open dialog box via File > Open, and locate the annotation files in the ...\Distribution\Server\Dynamics folder.
2. Select an annotation to view or modify.

   The Editor View tab appears, which can be used to edit and/or view the annotation.

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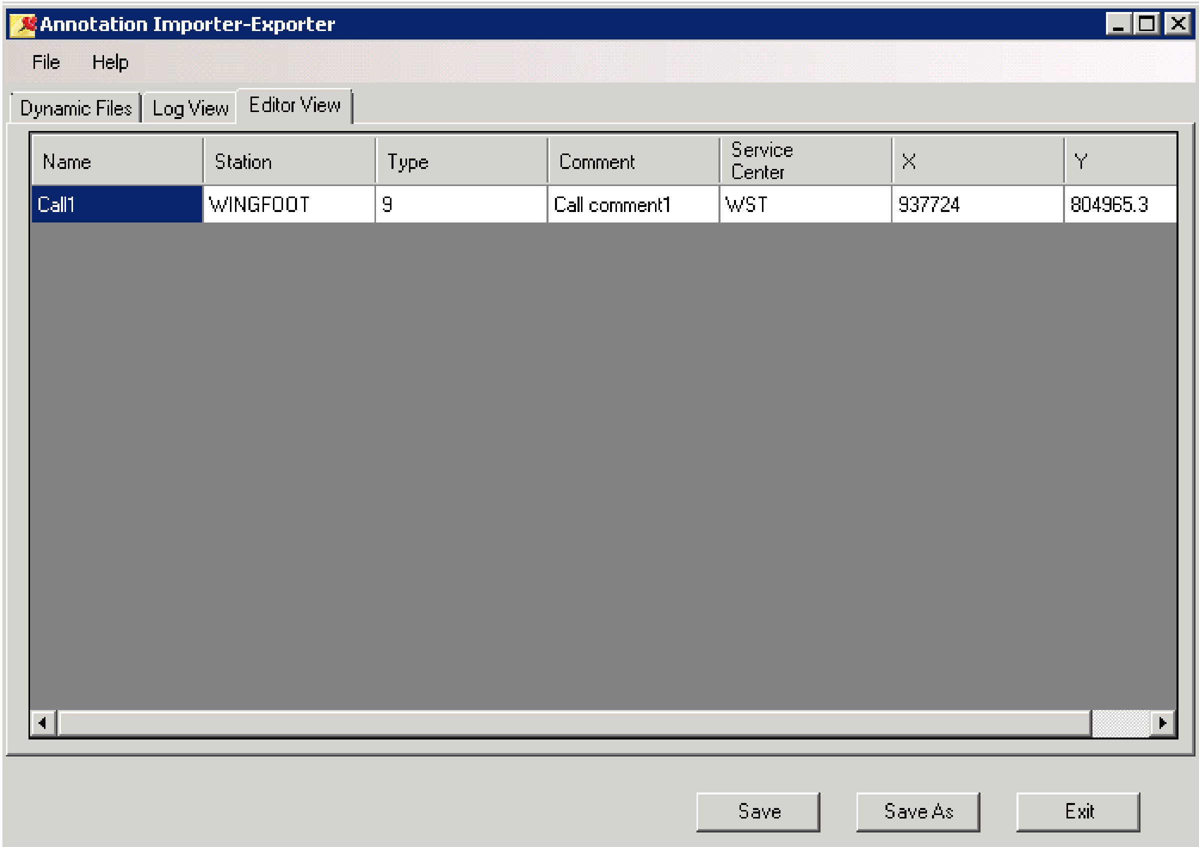


3. To save the edited annotation, click Save.
4. To update binary files with the changes in the XML files, navigate to the Dynamic Files tab and click the Import button for the corresponding file.

   The file's timestamp is updated, and an import log record is created. Changes are visible in the Geographic view.

5. To save the file under a different name, click Save As.

   The Save As dialog box opens.

6. Type in the new name, select the file type (binary or XML), and click Save.

2.5. Using the Command-Line Interface

Given a folder location with multiple SAV or XML files, you can use the command-line interface to import or export annotations in batches.

The following examples show how to perform sample command-line batch commands.

Sample batch import command:

```
AnnotationImporter.exe -IMPORT -BATCH -INPUT "C:\temp\RT" -OUTPUT "C:\temp\RT"
```

Sample batch export command:

```
AnnotationImporter.exe -EXPORT -BATCH -INPUT "C:\temp\RT" -OUTPUT "C:\temp\RT"
```

2.6. XML Schema Definition

The format for the annotation file is described in the XML Schema Definition (XSD) below, and illustrated in a sample XML file.

Multiple lines of text are supported by inserting a "new line" character (\n) in the comment field.

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema id="AnnotationCollection" targetNamespace="http://www.company.com"
xmlns:mstns="http://www.company.com" xmlns="http://www.company.com"
xmlns:xs="http://www.w3.org/2001/XMLSchema" xmlns:msdata="urn:schemas-microsoft-com:xml-msdata"
attributeFormDefault="qualified" elementFormDefault="qualified">
  <xs:element name="AnnotationCollection" msdata:IsDataSet="true" msdata:UseCurrentLocale="true">
    <xs:complexType>
      <xs:choice minOccurs="0" maxOccurs="unbounded">
        <xs:element name="Annotation">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="AnnotationType" type="xs:int" minOccurs="1" maxOccurs="1" />
              <xs:element name="Status" type="xs:int" minOccurs="1" maxOccurs="1" />
              <xs:element name="BitmapType" type="xs:int" minOccurs="0" maxOccurs="1" />
              <xs:element name="CreationTime" type="xs:dateTime" minOccurs="1" maxOccurs="1" />
              <xs:element name="UpdateTime" type="xs:dateTime" minOccurs="0" maxOccurs="1" />
              <xs:element name="Creator" type="xs:string" minOccurs="0" maxOccurs="1"  />
              <xs:element name="Updater" type="xs:string" minOccurs="0" maxOccurs="1"  />
              <xs:element name="Comment" type="xs:string" minOccurs="1" maxOccurs="1"  />
              <xs:element name="StatusText" type="xs:string" minOccurs="0" maxOccurs="1"/>
              <xs:element name="Name" type="xs:string" minOccurs="0" maxOccurs="1" />
              <xs:element name="Station" type="xs:string" minOccurs="0" maxOccurs="1" />
              <xs:element name="ServCtr" type="xs:string" minOccurs="0" maxOccurs="1" />
              <xs:element name="pt" minOccurs="1" maxOccurs="1">
                <xs:complexType>
                  <xs:sequence>
                    <xs:element name="X" type="xs:float" minOccurs="1" maxOccurs="1" />
                    <xs:element name="Y" type="xs:float" minOccurs="1" maxOccurs="1" />
                  </xs:sequence>
                </xs:complexType>
              </xs:element>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:choice>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

2.7. Sample XML File


```
<?xml version="1.0" encoding="utf-8"?>
<AnnotationCollection xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.company.com">
  <Annotation>
    <pt>
      <X>927096.6</X>
      <Y>813352.9</Y>
    </pt>
    <AnnotationType>0</AnnotationType>
    <Status>2</Status>
    <BitmapType>2</BitmapType>
    <CreationTime>2014-10-20T04:48:47-07:00</CreationTime>
    <UpdateTime>2014-10-20T04:48:47-07:00</UpdateTime>
    <Creator>DMSALL</Creator>
    <Updater />
    <DeviceLinks>DEVICES=Station=RAVEN,Device=17593613,DeviceType=,</DeviceLinks>
    <Geometry>DETAIL</Geometry>
    <StatusText>Annotation_School</StatusText>
    <Id>58db5004-26d2-40b9-a023-a9908f2fc2b6</Id>
    <ServCtr>WST</ServCtr>
    <Station>RAVEN</Station>
    <Comment>Comment1</Comment>
    <Name>KeyCustomer1</Name>
  </Annotation>
</AnnotationCollection>
```

3. Creating New Annotation Types

3.1. Defining New Annotation Types in the Server and Client

To define a new annotation type, perform the following steps:

1. Using Microsoft Windows Explorer, locate the `NetDynamicsConfigEditor.exe` file in the bin directory of the local installation.

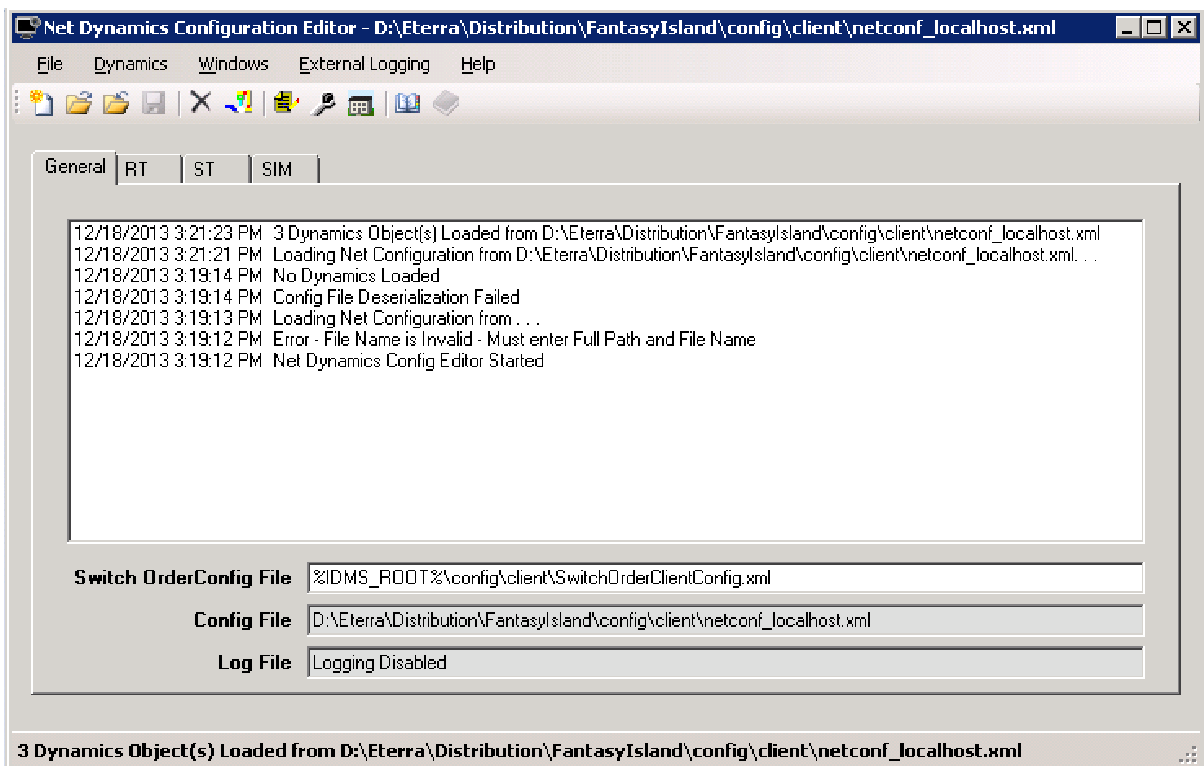
   The default installation path is:
   `D:\Eterra\Distribution\Configuration Tools\NetDynamicsConfigEditor.exe`

2. Double-click the `NetDynamicsConfigEditor.exe` file.

   The Net Dynamics Configuration Editor application starts.

3. To upload the network configuration file manually, select File > Open, and select the `netconf_localhost.xml` file (where <*localhost*> is a server name) from the list on the Open dialog box.

   The network configuration is uploaded, and three new tabs named "RT" (real-time), "ST" (study), and "SIM" (training simulator) appear, as shown below:

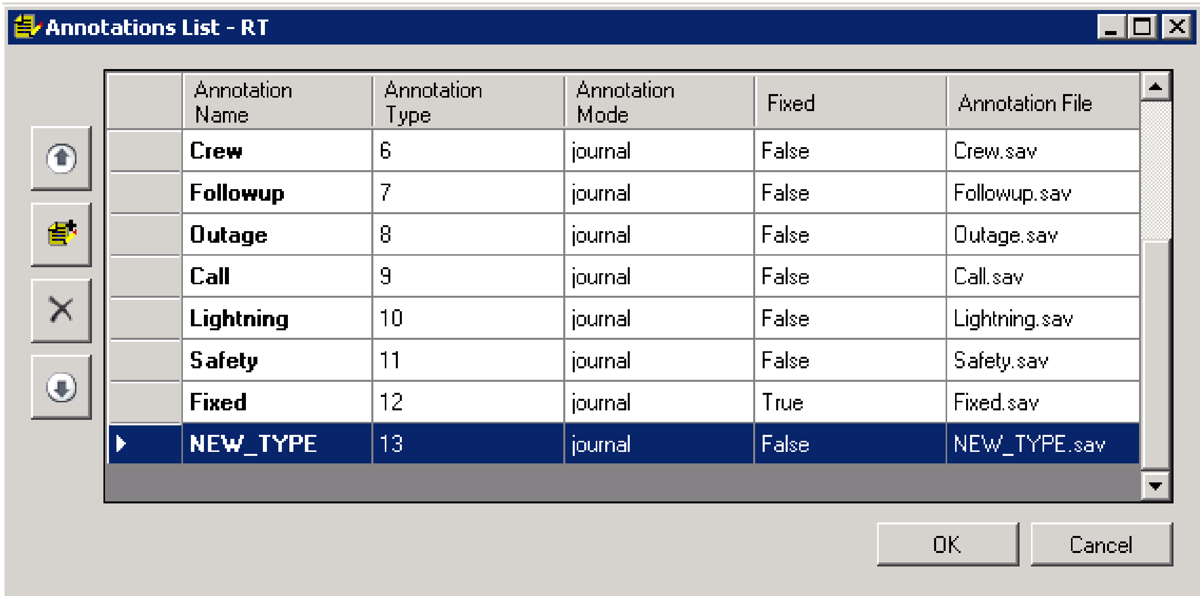


4. Select the RT tab, and click the Annotation button .

The Annotations List - RT pop-up is displayed.

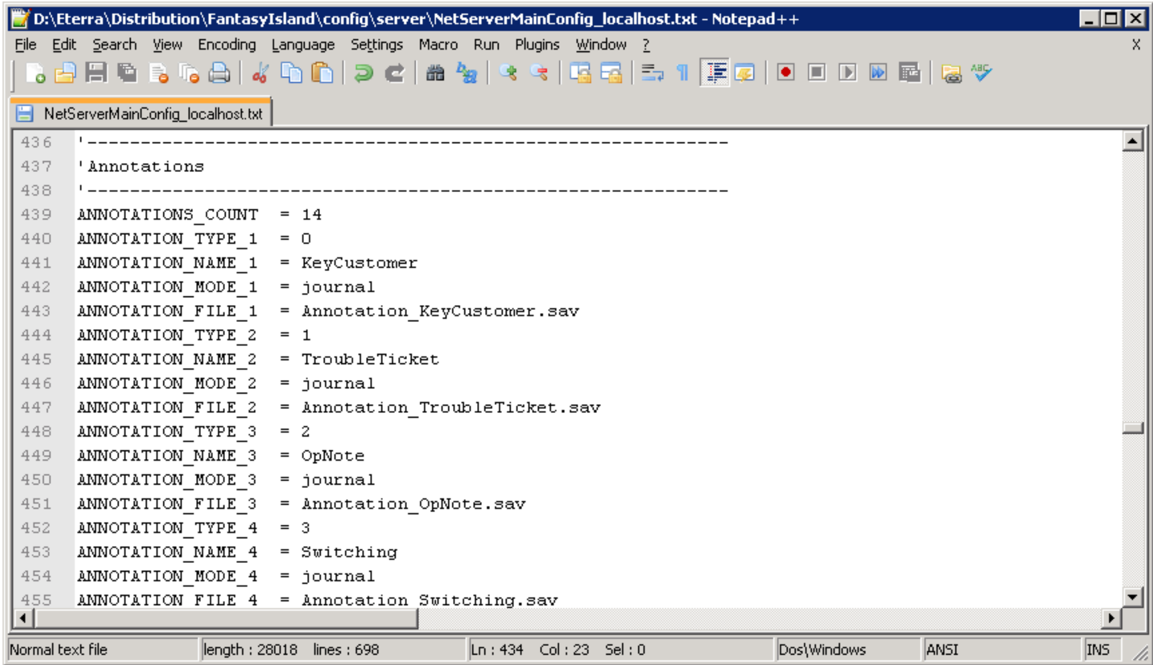
5. Click the Add Annotation Dynamics Entry button to the left of the Annotation grid section.

   A new blank entry appears in the Annotation list.



6. Enter the following fields:
   - **Annotation Name:** New annotation name.
   - **Annotation Type:** Sequential numeric value.
   - **Annotation Mode:** Journal.
   - **Fixed:** False.
   - **Annotation File:** Client annotation dynamics name (*.sav). Client RT annotations are stored in `%IDMS_ROOT%/dynamics/client/RT`.
7. Click OK.
8. Select the ST tab, and perform similar operations as described in steps 5-7.
9. Click File > Save to save the changes.
10. Open the RT server configuration file (e.g., `NetServerMainConfig.txt`).
11. Change the `ANNOTATION_COUNT` parameter according to the total number of annotations.
12. Add the following four parameters for the new annotation type:
    - **ANNOTATION_TYPE:** Annotation type specified in the Net Dynamics Configuration Editor.
    - **ANNOTATION_NAME:** Annotation name specified in the Net Dynamics Configuration Editor.
    - **ANNOTATION_MODE:** Annotation mode specified in the Net Dynamics Configuration Editor.
    - **ANNOTATION_FILE:** Server annotation dynamics name (*.sav). Client and server annotation

dynamics are stored in the `%IDMS_ROOT%/dynamics/server` folder.



13. Save the server configuration file.
14. Open the ST server configuration file (e.g., `NetServerStudyConfig.txt`), and perform similar operations as described in steps 11-13.
15. Restart the client and Net Server to implement the changes.

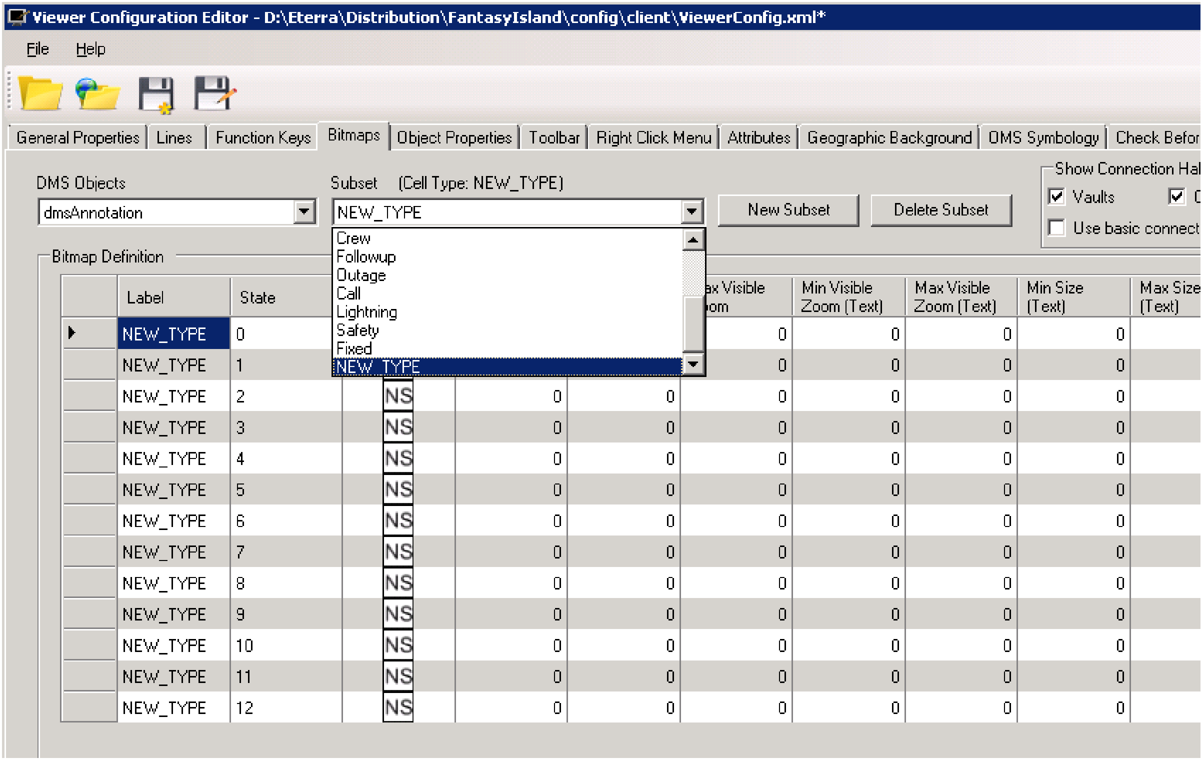
3.2. Configuring Visibility of a New Annotation Type

1. Run the Viewer Configuration Editor application.
2. Select the Bitmaps tab.
3. In the DMS Objects section, select the dmsAnnotation option.

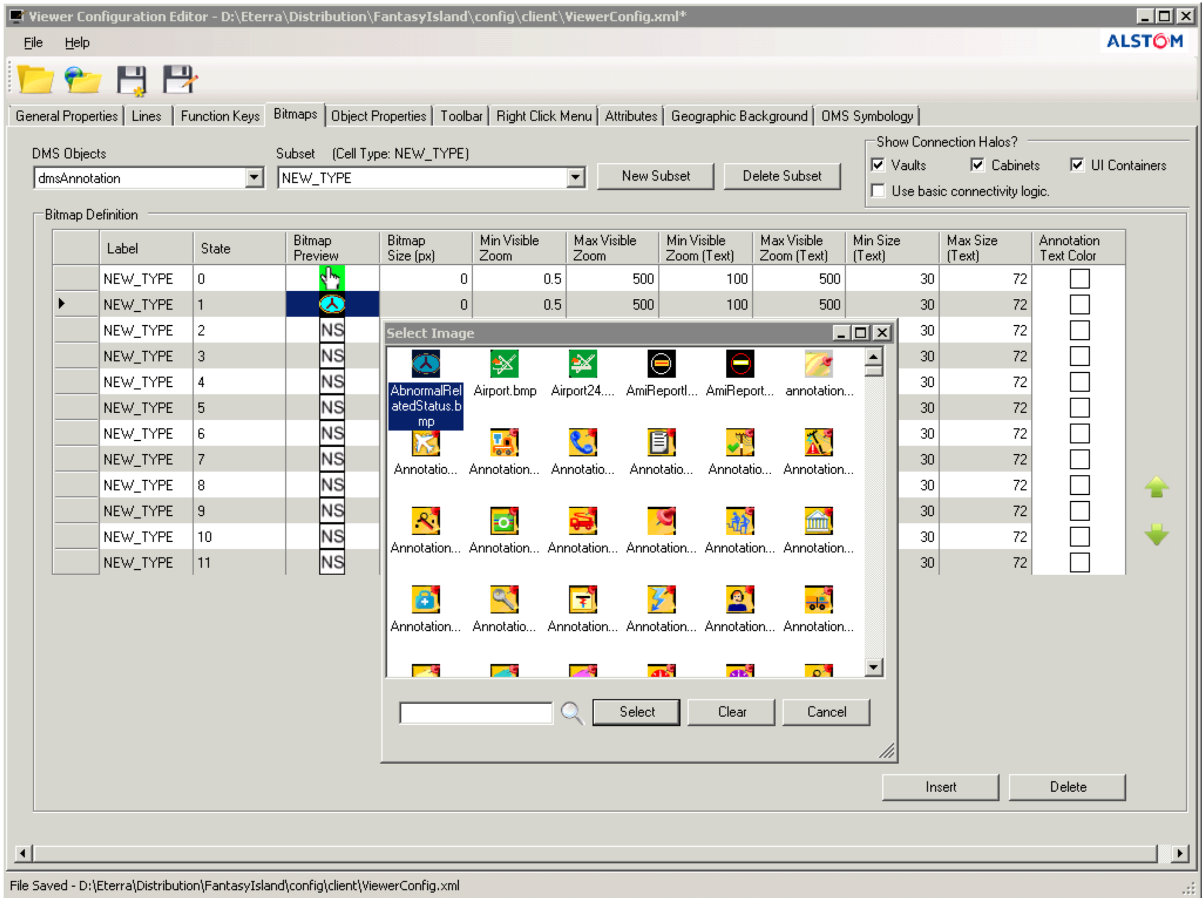
   In the Subset section, a list of objects from DNOM that is a subset of the Annotation DMS Object appears.
4. Click the New Subset button to the right of the Subset section.

   The Add Subset pop-up is displayed.
5. In the Name field, enter text identical to the text in the Net Dynamics Configuration Editor. Add the numeric value in the # States field, and then click OK.

   A new annotation entry is created at the bottom of the Subset list, as shown below.



6. Open the new Subset.
   - The Label field is automatically populated with the subset name. You can change the subset name by changing the label field.
   - The State List is a zero-based numeric list containing entries equal to the value entered in the # States field. If the # States field contains 3, the State List shows 0, 1, or 2.
7. For the new annotation type, enter values for the following fields: Size, Min Visible Zoom (Symbol), Max Visible Zoom (Symbol), Min Visible Zoom (Text), Max Visible Zoom (Text), Min Size (Text), and Max Size (Text).
   - Zero is a reasonable value that can be used for some of the fields.
   - The maximum font size is generally 72.
   - The maximum zoom size is 500.
8. Click the Bitmap Preview field to assign a bitmap to the selected annotation state.



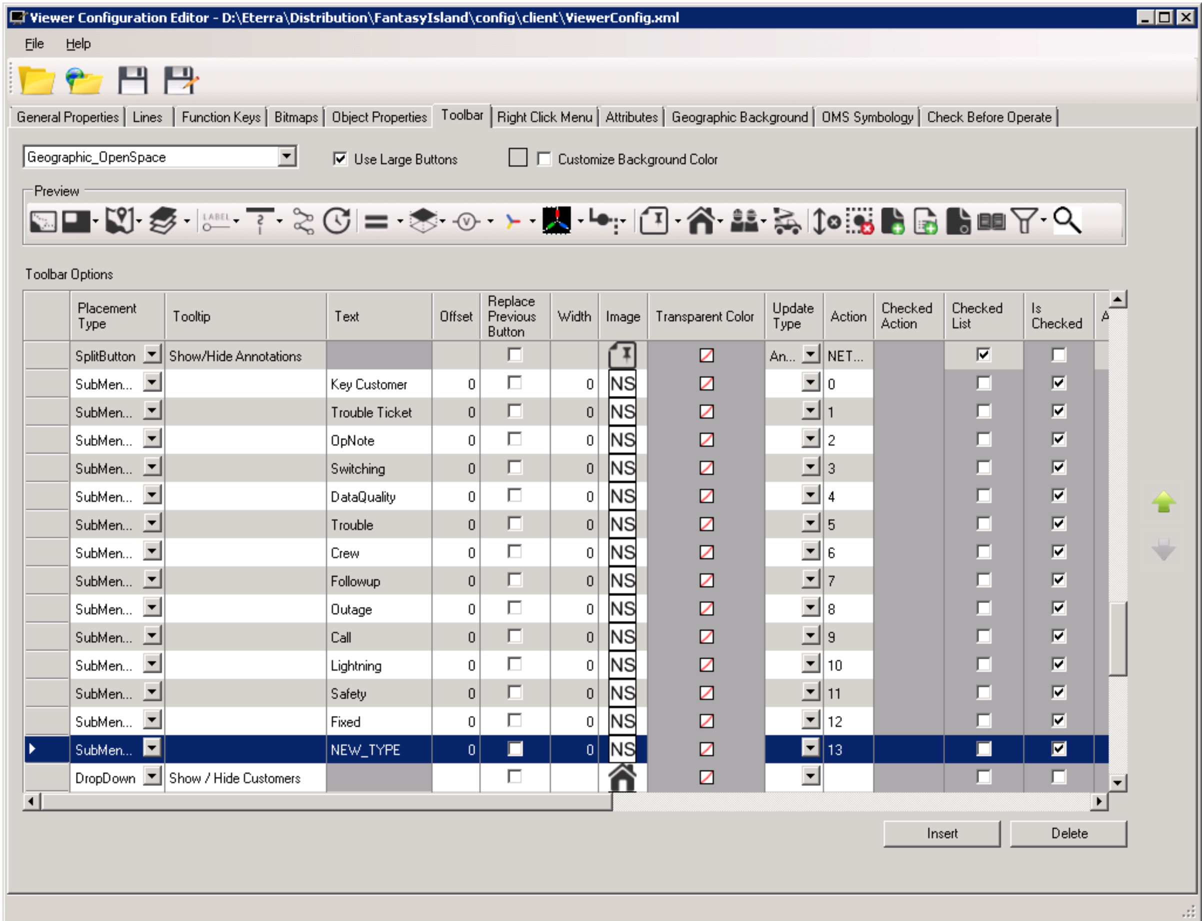
9. To save the changes, click File > Save.

3.3. Adding a New Annotation Type to the Geographic OpenSpace Toolbar

To add a new annotation type to the Annotation menu of the Geographic_Openspace (common) toolbar, perform the following steps:

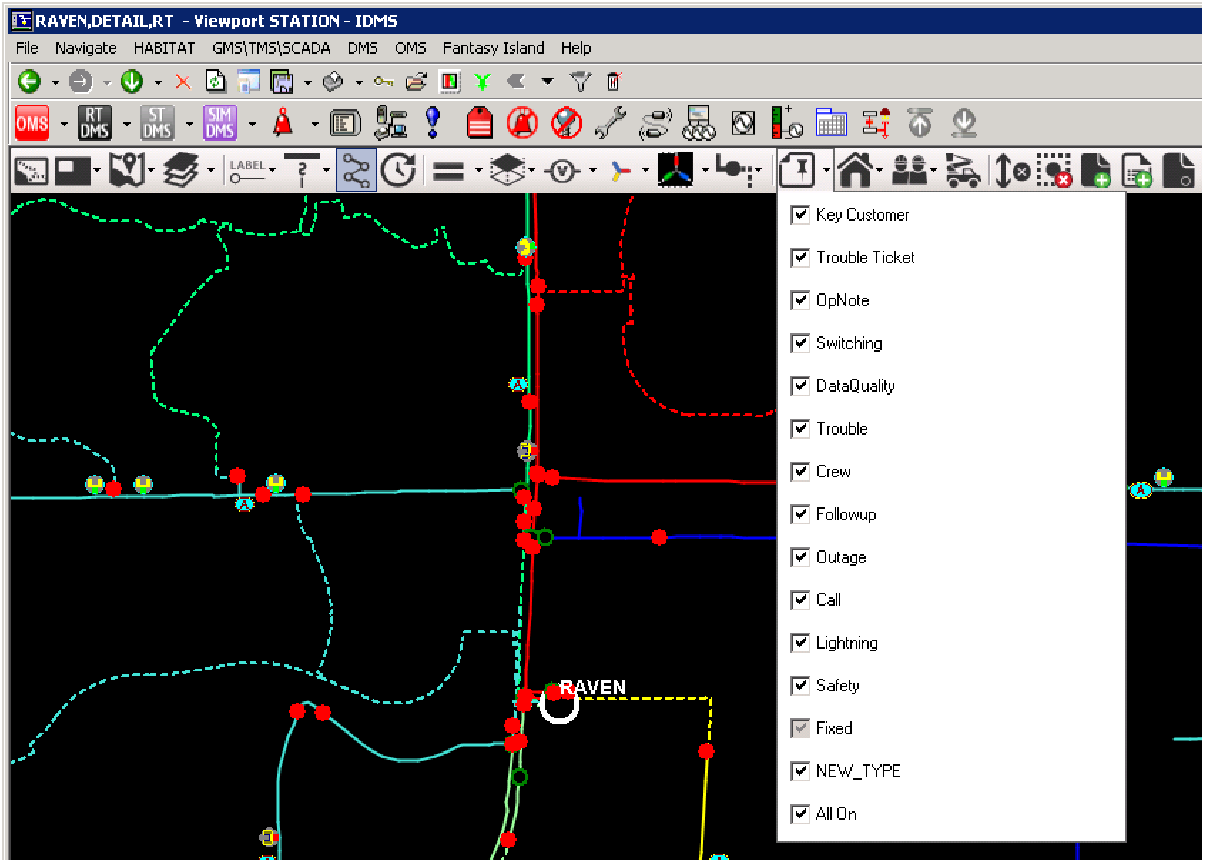
1. Run the Viewer Configuration Editor application.
2. In the Viewer Configuration Editor, select the Toolbar tab.
3. In the drop-down list, select Geographic_OpenSpace.
4. Click the Insert button to add a new row.

   The new row is inserted above the selected row. You can move each row to a new location using the up and down arrow buttons at the right of the Toolbar Options list.



5. For an annotation type identical to the new annotation type defined in the Net Dynamics Configuration Editor and the Viewer Configuration Editor, enter the following information (leave the remaining fields blank):
    - Offset = 0
    - Width = 0
    - Image = null
    - Text = Annotation type name
    - Action = Annotation type (numeric)
    - Is Checked = True
6. Select the new annotation row.
7. Using the up or down arrow button at the right of the Toolbar Options list, move the new annotation entry up to a location in the Geographic_Openspace toolbar grid below the entry for the Annotation button that appears on the toolbar.
8. To save the changes, select File > Save.

    The new annotation type will appear in the Annotations menu on the Geographic_OpenSpace (common) toolbar upon restarting the ADMS client.

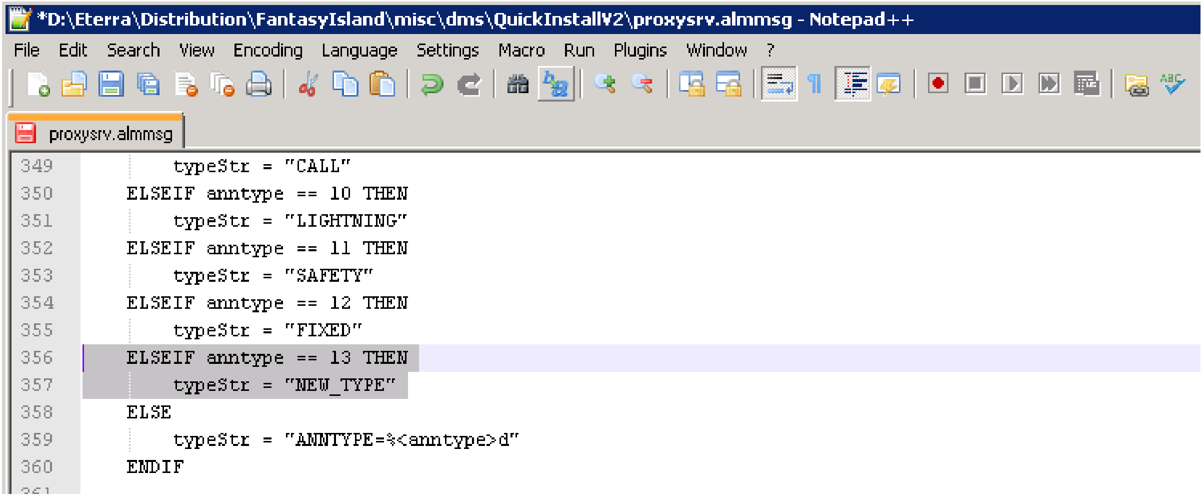


3.4. Updating the Alarms File

The ADMS product delivers the proxysrv.almmsg file, which includes alarm message definitions. It is necessary to update this file with the new annotation type.

1. Navigate to the proxysrv.almmsg file. The default location is %IDMS_ROOT\misc\dms\QuickInstallV2.
2. Open the file with a text editor.
3. Scroll to the "MSG dmsOp006 SEVERITY info" data block.
4. Insert a condition check for the new annotation type after the last entry:

```
ELSEIF anntype == 13 THEN
typeStr = "NEW_TYPE"
```



5. Save the proxysrv.almmsg file and copy it to the target folder (%HABUSER_SRCDIR%\scada\proxysrv) on the PDS server.
6. Compile the proxysrv.almmsg file on the PDS server:

```
> cd /d %HABUSER_SRCDIR%\scada\proxysrv
> buildlst -almmsg
```

   If the compilation is successful, the habalmmsg.dll in the %HABUSER_BINDIR% should be updated to include the updates for the NEW_TYPE annotation.

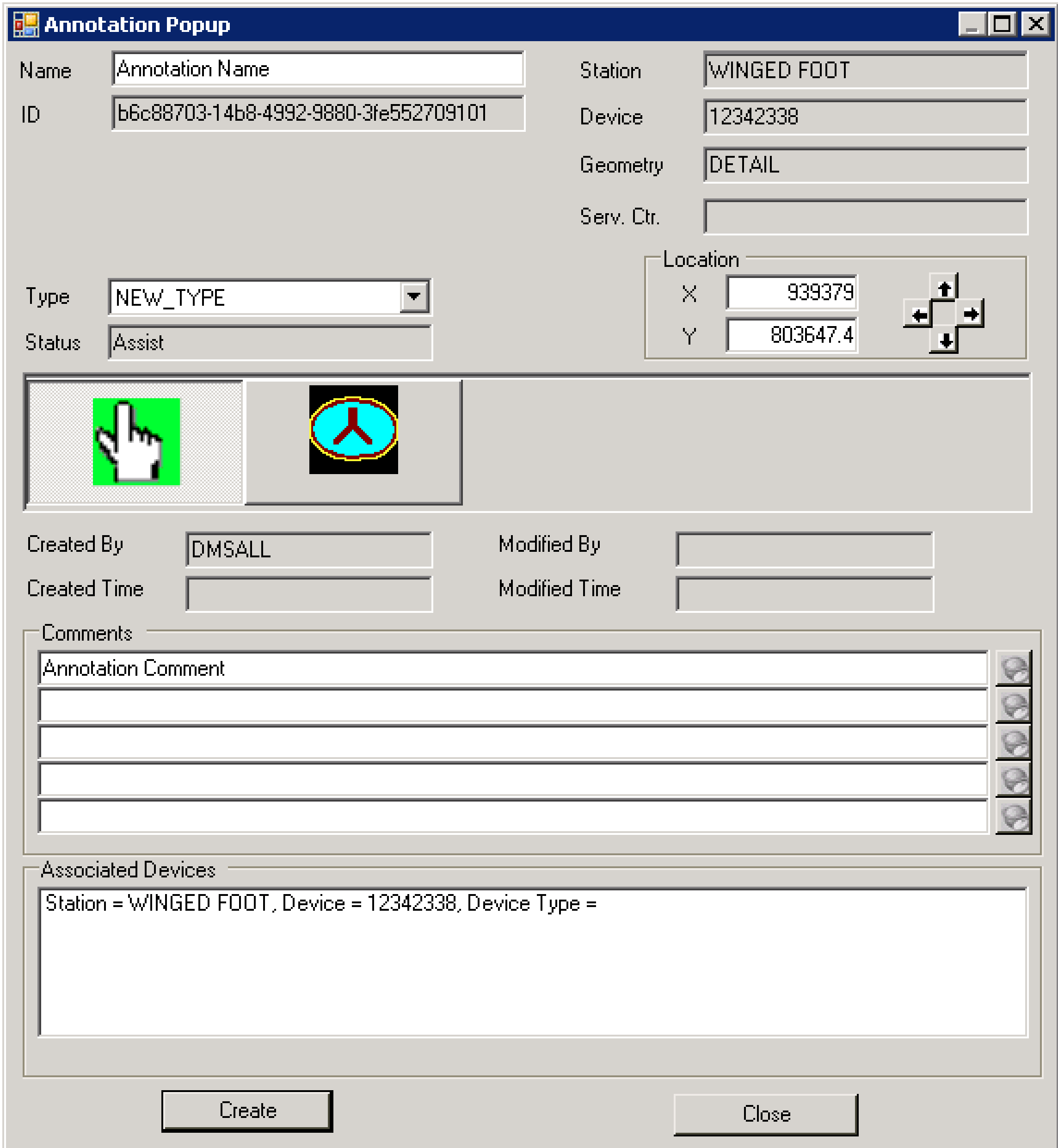
   Consult with a system administrator in case you need assistance in this step.

7. Copy habalmmsg.dll to the %HABUSER_BINDIR% folder on the real-time SCADA/EMS server.
8. Restart the SCADA/EMS system.

3.5. Placing Annotations

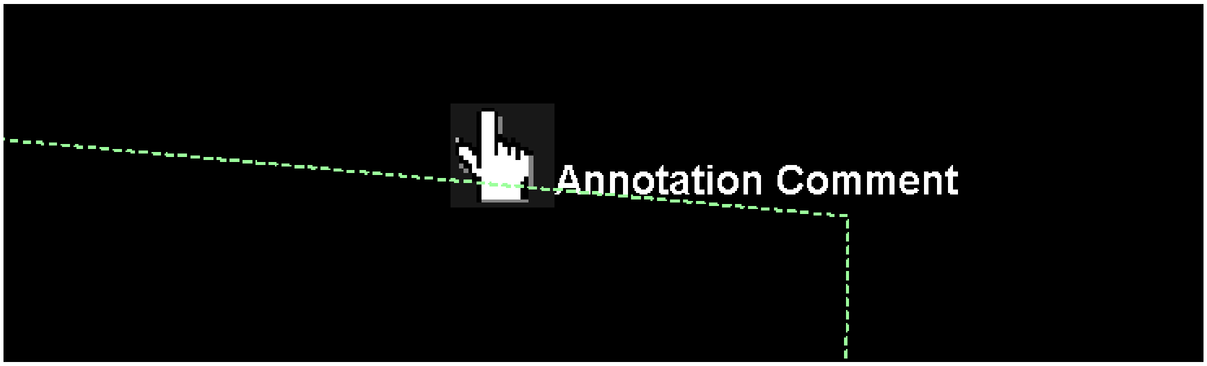
To place the new annotation, perform the following steps:

1. To call the Annotation pop-up, do one of the following:
   - Right-click on the geographic view and select the Place Annotation option from the context menu.
   - On the geographic view, press Ctrl+A.
2. Enter the annotation name. The maximum length is 18 characters.
3. Add a comment. The maximum length is 75 characters.
4. Edit the location and associated devices as necessary.
5. Select an annotation type from the Type list.



6. Click Create.

   The annotation is added at the specified location. To view the annotation and comment, set the zoom level as specified on the Bitmaps tab of the Viewer Configuration Editor.



When annotations are added, updated or deleted, the operations are logged in the System Activity Log, which can be accessed from the Main toolbar.